Deep Learning, Advanced Course, DD2412

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Working title: Ranking Unsupervised Pre-Training

Description

In this group project we are aiming to reproduce the paper *RankMe: Assessing the Downstream Performance of Pretrained Self-Supervised Representations by Their Rank* and all of its results on the same datasets as used in the paper. Furthermore, we are planning to evaluate the paper on machine learning research practices with a focus on reproducability. We are aiming for the *Good-Very Good* grade range, however we would already like to discuss a possible upgrade to the *Excellent* range.

Milestones

- Implement the unsupervised pre-training algorithms of the paper with the following three different objectives: SimCLR, VICReg, and DINO. We are skipping the two VICReg variants due to computational constraints.
- Perform a small scale training run to validate our implementation.
- Perform the training of the paper on a subset of imagenet https://github.com/fastai/imagenette with a simplified ResNet-18.
- Evaluate performance of the pretrained models on the training data (imagenet subset), iNaturalist18 and SUN397 to compare performance on the pretrained dataset and two new datasets. These datasets specifically are chosen, as they are also the ones presented in the paper presentation.
- Compare different training objectives and their respective performances.
- Compare trend of performances to original paper.
- Experiment 1: Track RankME value over training time and evaluate robustness and speed of convergence.

- Experiment 2: Train linear and two layer heads for finetuning and compare accuracies.
- Experiment 3: If feasible in given time: Adversarial one-pixel attack to evaluate model robustness.
- Experiment 4: If feasible in given time: Train different projection heads and compare Rank convergence and compare capacity for finetuning.
- Experiment 5: Check RankMe's sensitivity to the hyperparameters and to the design choices by additional experiments that we will carry out where we will try different hyperparameters and evaluate their impact on the outcome. We will do random search for determining the hyperparameters to test as it is expected to be more computationally efficient despite entailing higher variance. Furthermore, in terms of design choices, we will use ResNet-18 architecture instead of ResNet-50 and see how a simpler network affects performance compared to the ones presented in the paper.
- Evaluate original paper on the Clarity of: Method Description, Algorithm Pseudocode, Mathematical Formulations, and Data Description
- Evaluate original paper on Transparency, Discussion of Alternatives, Generalizability, Citation Quality, and potential Ethical Considerations

Implementation Details

We will need a lot of computational resources for pretraining all those models. Therefore, we are planning to make use of the Google Cloud Credits given to us. We expect that the combined number of credits should be more than sufficient.

We are planning on using *Pytorch* as a framework for this project and we will measure the success of our project based on whether we will be able to reproduce the vast amount of results of the original paper.

Skills we aim to acquire

- *Markus:* Get insights into unsupervised training objectives and the how to evaluate potential performance of pretrained models without using labels. The capacity to implement arbitrary training objectives just by reading about them in a paper.
- *Caner:* Acquire the capability to proficiently implement research papers in the domain of Self-Supervised Learning (SSL),particularly with a focus on utilizing large datasets and the Jax framework. In-Depth Understanding of SSL Methods: Develop a profound comprehension of VICReg, its various derivatives, and DINO by studying their theoretical foundations and practically implementing throughout the project.
- *Sruti:* Intending to develop a detailed understanding of unsupervised training, self-supervised learning and supervised evaluation. Study im-

pact of hyperparameters on model performance and evaluation. Aiming to gain skills for critical analysis and performance comparison. Will study working on evaluation and comparing our results with the original paper, identifying key areas for revision and improvement.

Grade Justification

We are planning on completely reproducing the whole paper and furthermore evaluate the paper on its reproducability. Furthermore we introduce several reproducability experiments. Therefore, we are going above the original task of simply reproducing the paper and feel that the *Good-Very Good* range would be adequate if we implement Experiment 1,2 and 5. Furthermore, if we happen to have more time than expected we would like to implement Experiment 3 and 4. As this is going above and beyond reproducability, we feel like this should be a goal for the *Excellent* range.